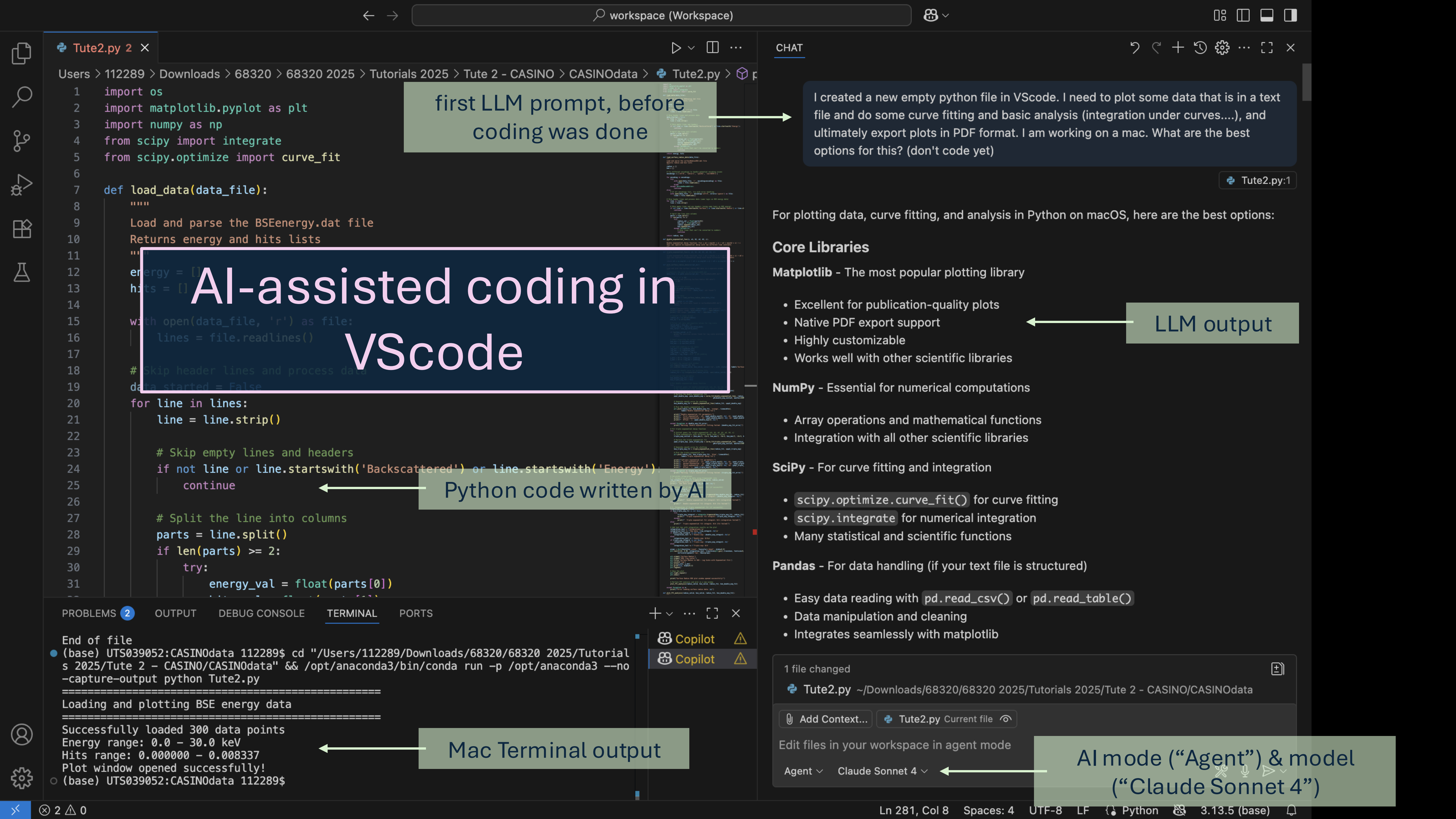
68320 Nanofabrication and and Nanocharacterization Techniques

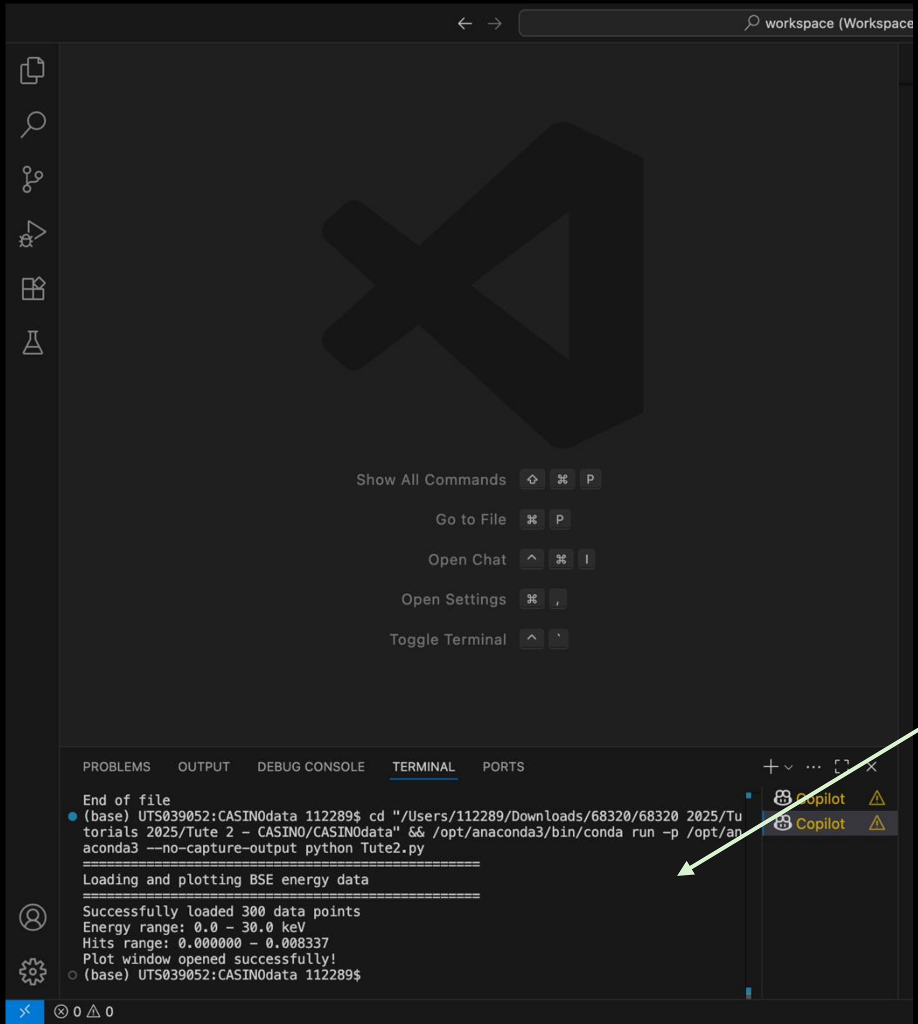
Tutorial 2 cheat-sheet: How to use AI to help you in this subject



AI-assisted coding in VScode
first LLM prompt, before coding was done
LLM output
Python code written by AI
Mac Terminal output
AI mode ("Agent") & model ("Claude Sonnet 4")
workspace (Workspace)
Tute2.py 2
Users > 112289 > Downloads > 68320 > 68320 2025 > Tutorials 2025 > Tute 2 - CASINO > CASINOdata > Tute2.py
import os
import matplotlib.pyplot as plt
import numpy as np
from scipy import integrate
from scipy.optimize import curve_fit
def load_data(data_file):
Load and parse the BSEenergy.dat file
Returns energy and hits lists
for line in lines:
line = line.strip()
# Skip empty lines and headers
if not line or line.startswith('Backscattered') or line.startswith('Energy'):
continue
# Split the line into columns
parts = line.split()
if len(parts) >= 2:
try:
energy_val = float(parts[0])
PROBLEMS 2
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
End of file
(base) UTS039052:CASINOdata 112289$ cd "/Users/112289/Downloads/68320/68320 2025/Tutorials 2025/Tute 2 - CASINO/CASINOdata" && /opt/anaconda3/bin/conda run -p /opt/anaconda3 --no-capture-output python Tute2.py
Loading and plotting BSE energy data
Successfully loaded 300 data points
Energy range: 0.0 - 30.0 keV
Hits range: 0.000000 - 0.008337
Plot window opened successfully!
(base) UTS039052:CASINOdata 112289$
Copilot
Copilot
CHAT
I created a new empty python file in VScode. I need to plot some data that is in a text file and do some curve fitting and basic analysis (integration under curves....), and ultimately export plots in PDF format. I am working on a mac. What are the best options for this? (don't code yet)
Tute2.py:1
For plotting data, curve fitting, and analysis in Python on macOS, here are the best options:
Core Libraries
Matplotlib - The most popular plotting library
Excellent for publication-quality plots
Native PDF export support
Highly customizable
Works well with other scientific libraries
NumPy - Essential for numerical computations
Array operations and mathematical functions
Integration with all other scientific libraries
SciPy - For curve fitting and integration
scipy.optimize.curve_fit() for curve fitting
scipy.integrate for numerical integration
Many statistical and scientific functions
Pandas - For data handling (if your text file is structured)
Easy data reading with pd.read_csv() or pd.read_table()
Data manipulation and cleaning
Integrates seamlessly with matplotlib
1 file changed
Tute2.py ~/Downloads/68320/68320 2025/Tutorials 2025/Tute 2 - CASINO/CASINOdata
Add Context...
Tute2.py Current file
Edit files in your workspace in agent mode
Agent
Claude Sonnet 4
Ln 281, Col 8
Spaces: 4
UTF-8
LF
Python
3.13.5 (base)

End of LLM output to my 1st prompt

My 2nd prompt



Recommended Approach

Start with **matplotlib + numpy + scipy** as your core stack. Add **pandas** if your data is in a structured format (CSV-like). This combination will handle all your requirements efficiently and is the industry standard for scientific Python work.

Would you like me to help you set up the environment and start coding with these libraries?

> Yes. First, can you write code that opens a console window, opens the file BSEenergy.dat (located in the same directory as the code Tute2.py) and displays the file contents?

Tute2.py:1

I'll help you write code to open and display the contents of the BSEenergy.dat file. First, let me configure the Python environment and then write the code.

> Configuring a Python Environment

Now I'll write the code to open and display the BSEenergy.dat file:

Tute2.py +44 -1

Now let's run the code to see the contents of the BSEenergy.dat file:

```
cd "/Users/112289/Downloads/68320/68320 2025/Tutorials 2025/Tute 2 - CASINO/CASINOdata" && /opt/anaconda3/bin/conda run -p /opt/anaconda3 --no-capture-output python Tute2.py
```

Perfect! The code successfully opened and displayed the contents of the BSEenergy.dat file.

Here's what we can see from the data:

1. **File Structure**: The file contains backscattered energy data with two columns:

1 file changed

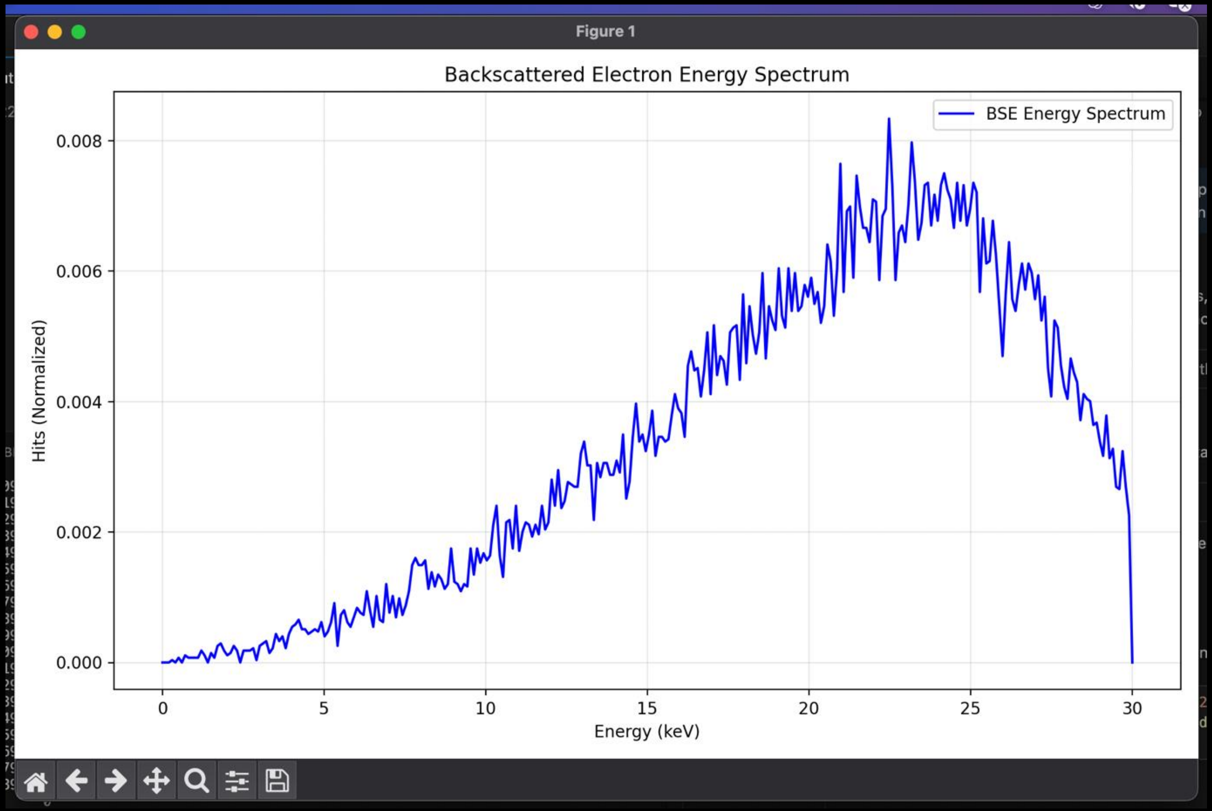
Tute2.py ~/Downloads/68320/68320 2025/Tutorials 2025/Tute 2 - CASINO/CASINOdata

Add Context...

Edit files in your workspace in agent mode

Agent Claude Sonnet 4

My next prompt was to import the data in the file and plot it…

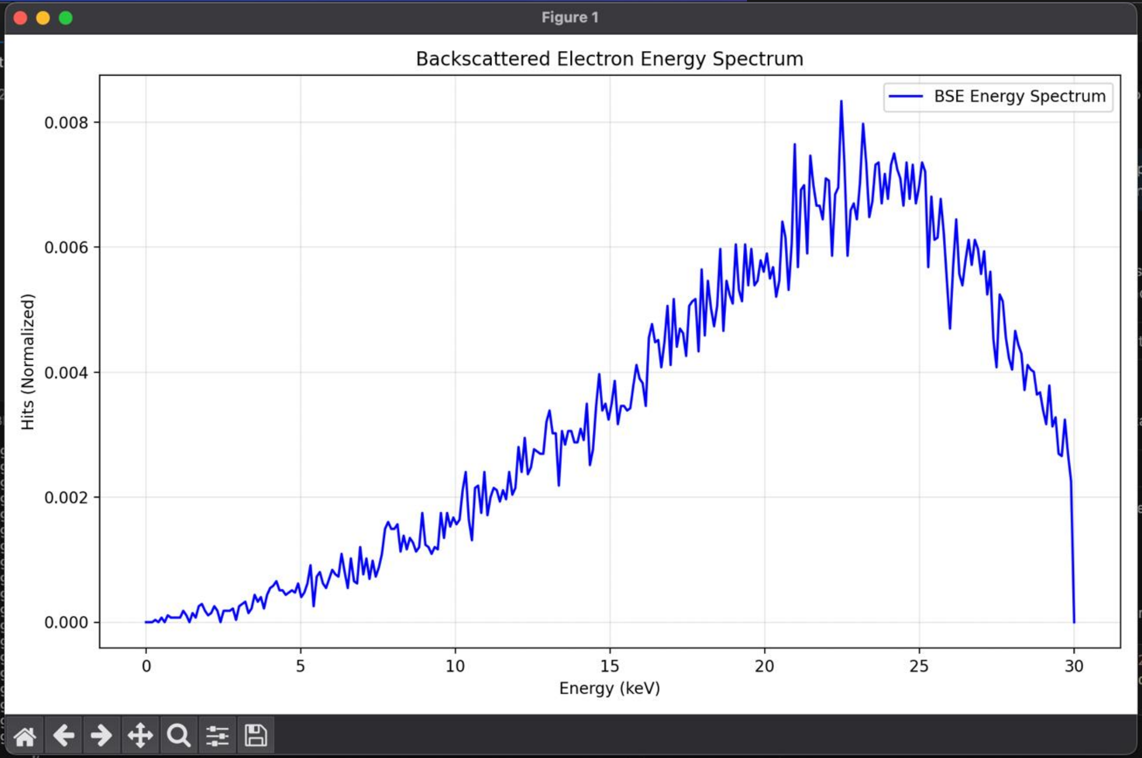


Is the plot window visible on your screen? Would you like me to add any additional features like curve fitting or data analysis next?

Yes, I want to display the data as points (not a line) and fit a curve to it - perhaps try 2nd, 3rd and 4th order polynomial fits and display them all on the plot.

Tute2.py:1-14

python



parse this data and create plots using matplotlib?

plays the file contents and instead import the data
n a window and plot the data in it

Tute2.py:44

s, parse the data into lists suitable for plotting, and
ckages.

thon 3.13.5 (base)

ails:

eady installed in the environment. Now let me modify

nd create the plot:

```
20 2025/Tutorials 2025/Tute 2 - CASINO/
da run -p /opt/anaconda3
```

```
End of file
(base) UTS039052:CASINOdata 112289$ cd "/Users/112289/Downloads/68320/68320 2025/Tutorials 2025/Tute 2 - CASINO/CASINOdata" && /opt/anaconda3/bin/conda run -p /opt/anaconda3 --no-capture-output python Tute2.py
==================================================
Loading and plotting BSE energy data
==================================================
Successfully loaded 300 data points
Energy range: 0.0 - 30.0 keV
Hits range: 0.000000 - 0.008337
```

1 file changed Keep Undo

Tute2.py ~/Downloads/68320/68320 2025/Tutorials 2025/Tute 2 - CASINO/CASINOdata

Add Context... Tute2.py Current file

Edit files in your workspace in agent mode

Agent Claude Sonnet 4

Ln 89, Col 11 Spaces: 4 UTF-8 {} Python 3.13.5 (base)

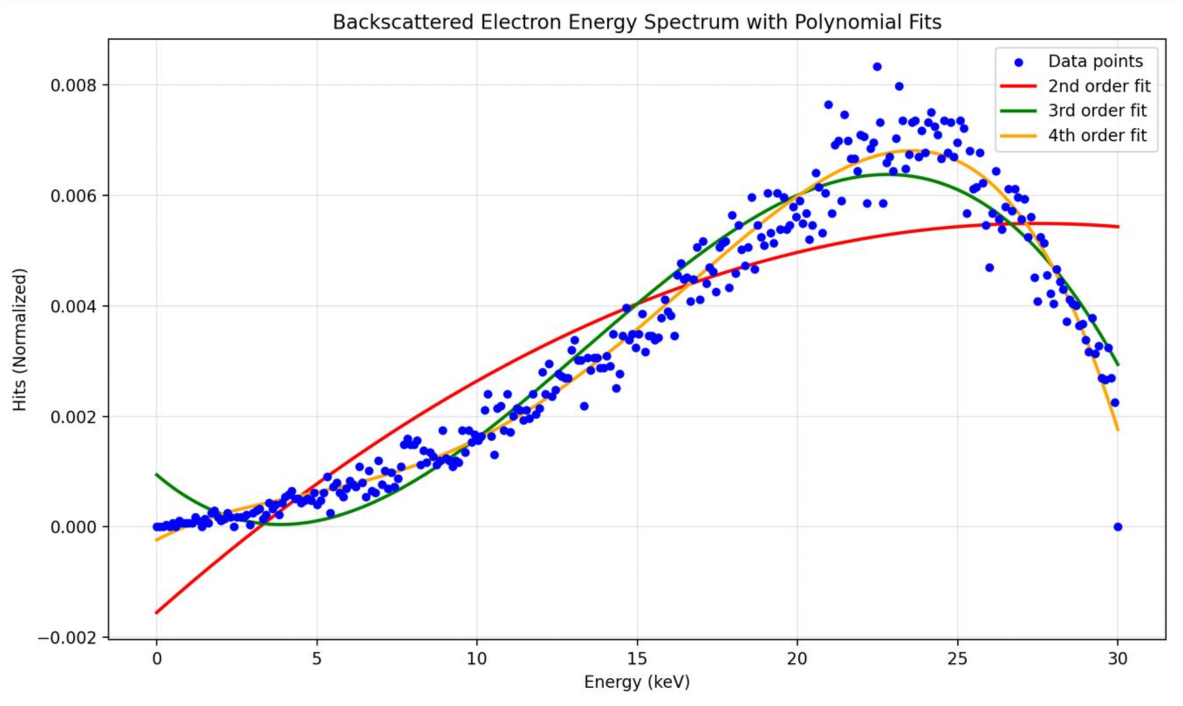
You can now visually compare the data to the different polynomial fits. Let me know if you want to see fit coefficients, residuals, or further analysis!

Change the color of the 2nd and 3rd order polynomial fits by making it 50% brighter (in RGB space), so they appear as faint on the plot.

Tute2.py:65-78

python Thu 7 Aug 8:34 AM

Figure 1



not a line) and fit a curve to it - perhaps try 2nd,
play them all on the plot.

Tute2.py:1-14

20 2025/Tutorials 2025/Tute 2 - CASINO/
da run -p /opt/anaconda3

1 file changed Keep Undo

Tute2.py ~/Downloads/68320/68320 2025/Tutorials 2025/Tute 2 - CASINO/CASINOdata

Add Context... Tute2.py Current file

Edit files in your workspace in agent mode

Agent Claude Sonnet 4

Ln 5, Col 26 Spaces: 4 UTF-8 Python 3.13.5 (base)

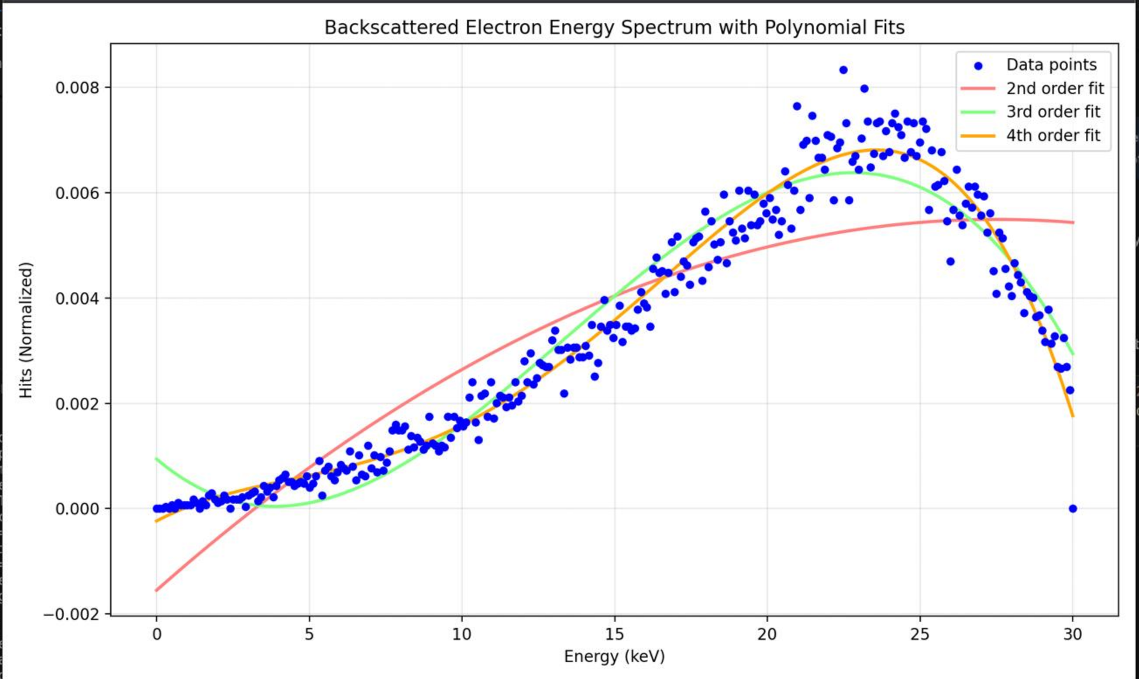
The 2nd and 3rd order polynomial fits now appear much fainter on the plot, making it easier to focus on the data points and the 4th order fit while still being able to see the comparison between different polynomial orders. The lighter colors were achieved by mixing the original red and green with white in RGB space to create the 50% brighter versions.

Integrate the raw data, and also the 4th order poly fit & display both results on the graph

Tute2.py:73-86

python — Thu 7 Aug 8:53 AM

Figure 1



r polynomial fits by making it 50% brighter (in RGB
t.

Tute2.py:65-78

ynomial fits to make them 50% brighter (lighter) so

th the lighter 2nd and 3rd order polynomial fits:

```
20 2025/Tutorials 2025/Tute 2 - CASINO/
da run -p /opt/anaconda3
```

```
Successfully loaded 300 data points
Energy range: 0.0 - 30.0 keV
Hits range: 0.000000 - 0.008337
```

1 file changed — Keep — Undo

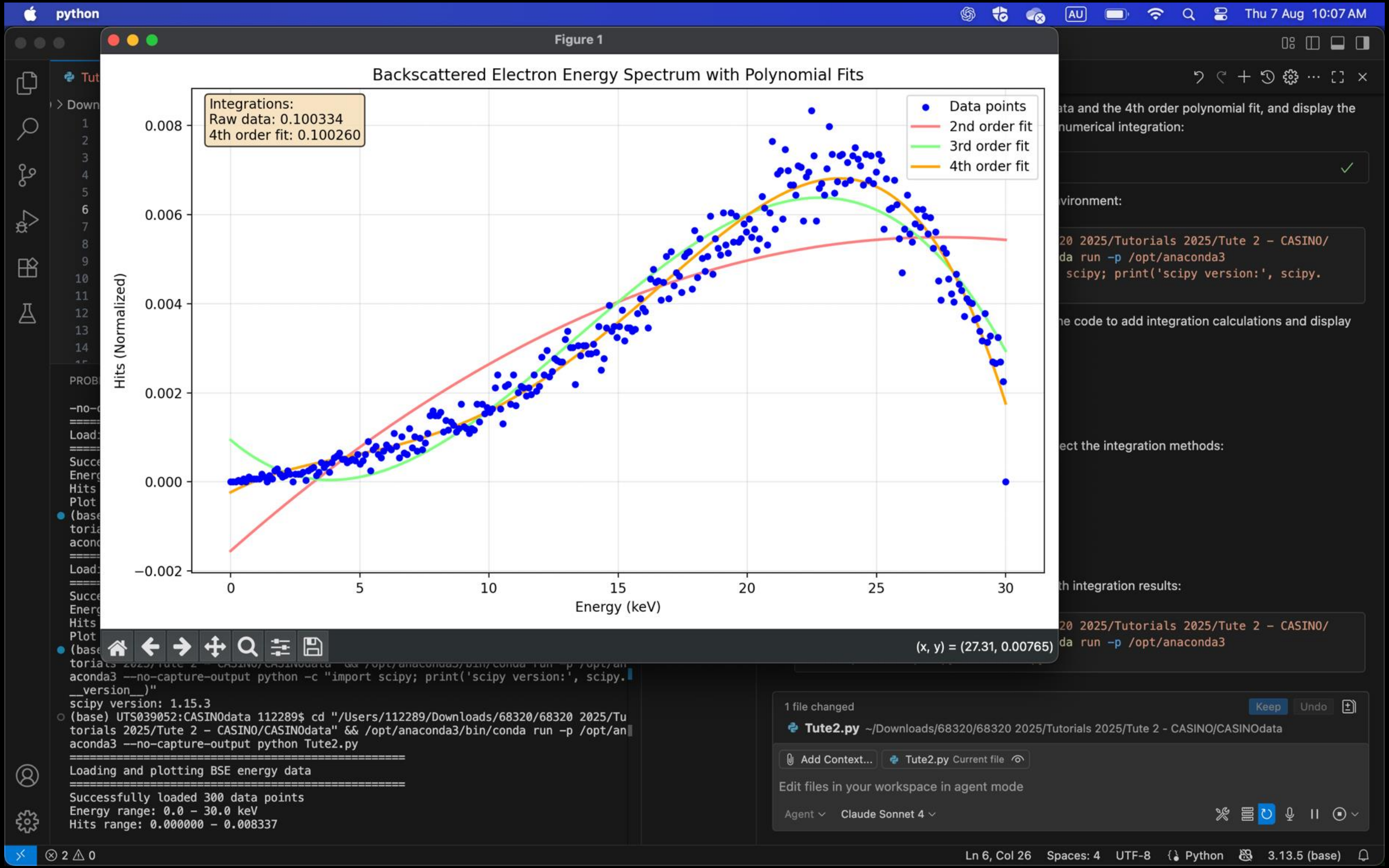
Tute2.py ~/Downloads/68320/68320 2025/Tutorials 2025/Tute 2 - CASINO/CASINOdata

Add Context... — Tute2.py Current file

Edit files in your workspace in agent mode

Agent — Claude Sonnet 4

Ln 5, Col 26 — Spaces: 4 — UTF-8 — Python — 3.13.5 (base)


python
Thu 7 Aug 10:07 AM
Figure 1
Backscattered Electron Energy Spectrum with Polynomial Fits
Integrations:
Raw data: 0.100334
4th order fit: 0.100260
Data points
2nd order fit
3rd order fit
4th order fit
Hits (Normalized)
Energy (keV)
(x, y) = (27.31, 0.00765)
ata and the 4th order polynomial fit, and display the
numerical integration:
vironment:
20 2025/Tutorials 2025/Tute 2 - CASINO/
da run -p /opt/anaconda3
scipy; print('scipy version:', scipy.
he code to add integration calculations and display
ect the integration methods:
th integration results:
20 2025/Tutorials 2025/Tute 2 - CASINO/
da run -p /opt/anaconda3
aconda3 --no-capture-output python -c "import scipy; print('scipy version:', scipy.
__version__)"
scipy version: 1.15.3
(base) UTS039052:CASINOdata 112289$ cd "/Users/112289/Downloads/68320/68320 2025/Tu
torials 2025/Tute 2 - CASINO/CASINOdata" && /opt/anaconda3/bin/conda run -p /opt/an
aconda3 --no-capture-output python Tute2.py
Loading and plotting BSE energy data
Successfully loaded 300 data points
Energy range: 0.0 - 30.0 keV
Hits range: 0.000000 - 0.008337
1 file changed
Keep
Undo
Tute2.py ~/Downloads/68320/68320 2025/Tutorials 2025/Tute 2 - CASINO/CASINOdata
Add Context...
Tute2.py Current file
Edit files in your workspace in agent mode
Agent
Claude Sonnet 4
Ln 6, Col 26
Spaces: 4
UTF-8
Python
3.13.5 (base)

The plot window should now be displaying both the visual fits and the quantitative integration results!

Leave the above code as is, and add code that does the following: (1) import data in the file surfaceRadiusBSE.dat, (2) plot it in a second window and make the y-axis log scale (set the min & max values on the y-axis using a smart algo that detects the minimum and maximum values in the dataset). Do not do any curve fitting or calculations yet.

Tute2.py:1-23

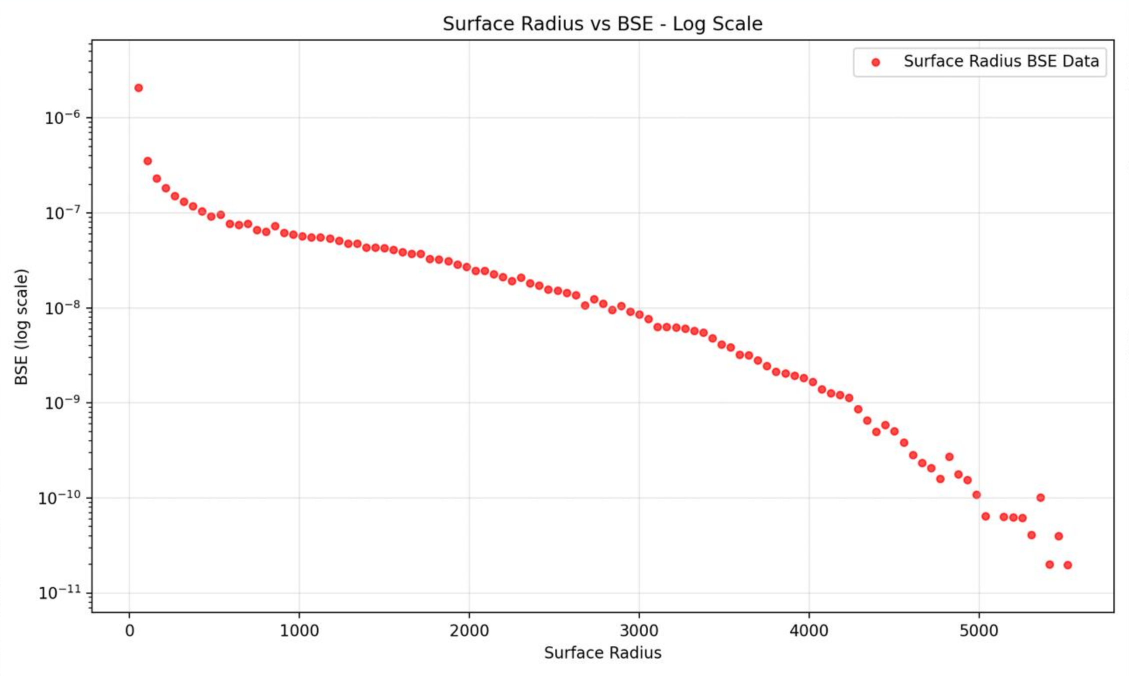
python Thu 7 Aug 10:16 AM

workspace (Workspace)

Tute2.py

CHAT

Figure 1



radius data: 'utf-8' codec can't decode byte 0xb2
hat the data format in surfaceRadiusBSE.dat is the
mport algo should work

Tute2.py:35-57

e might be in a different encoding. Let me modify the
different encodings, similar to how the BSE energy

ndle the encoding issue:

```
320 2025/Tutorials 2025/Tute 2 - CASINO/
nda run -p /opt/anaconda3
```

Keep Undo

Tute2.py ~/Downloads/68320/68320 2025/Tutorials 2025/Tute 2 - CASINO/CASINOdata

Add Context... Tute2.py Current file

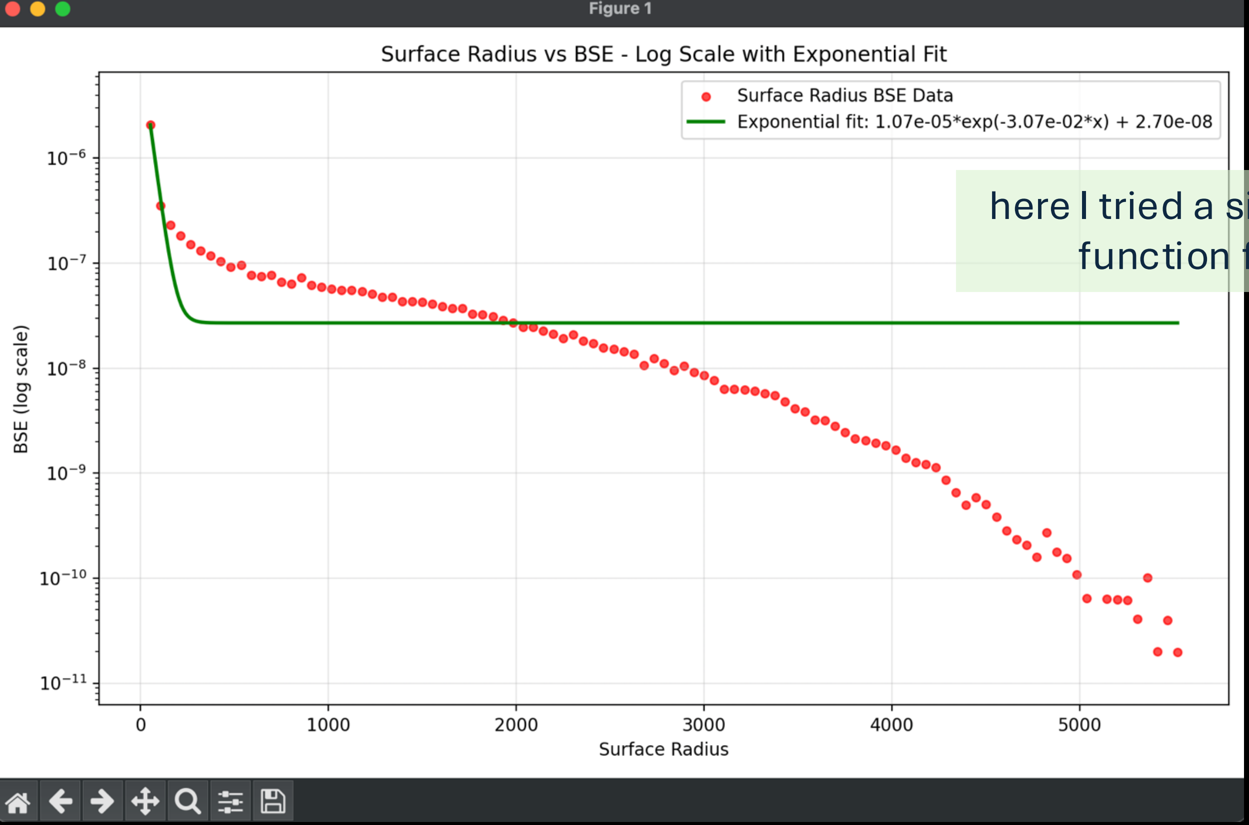
Edit files in your workspace in agent mode

Agent Claude Sonnet 4

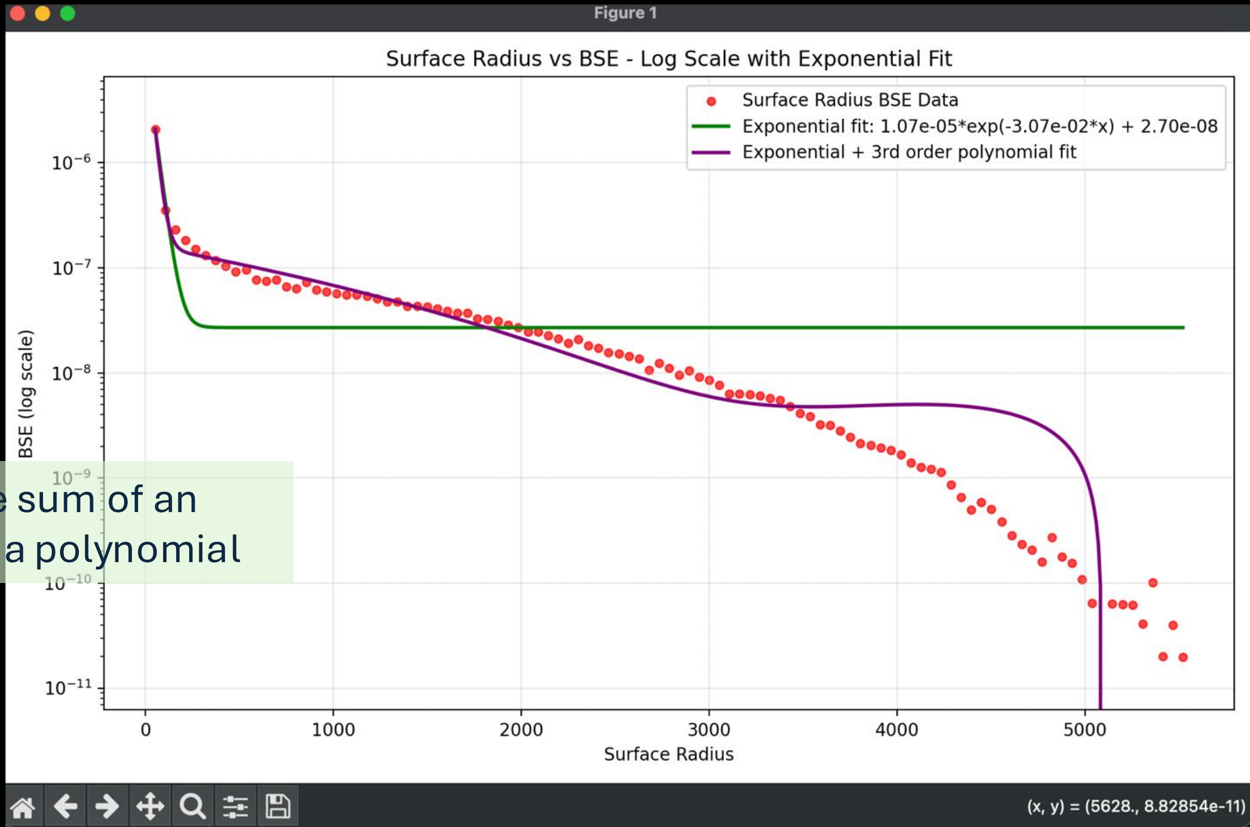
```
Plot window opened successfully!

==================================================
Loading and plotting Surface Radius BSE data
==================================================
Successfully loaded 300 data points
Radius range: 0.000000 - 16019.300000
BSE range: 0.000000e+00 - 2.087230e-06
```

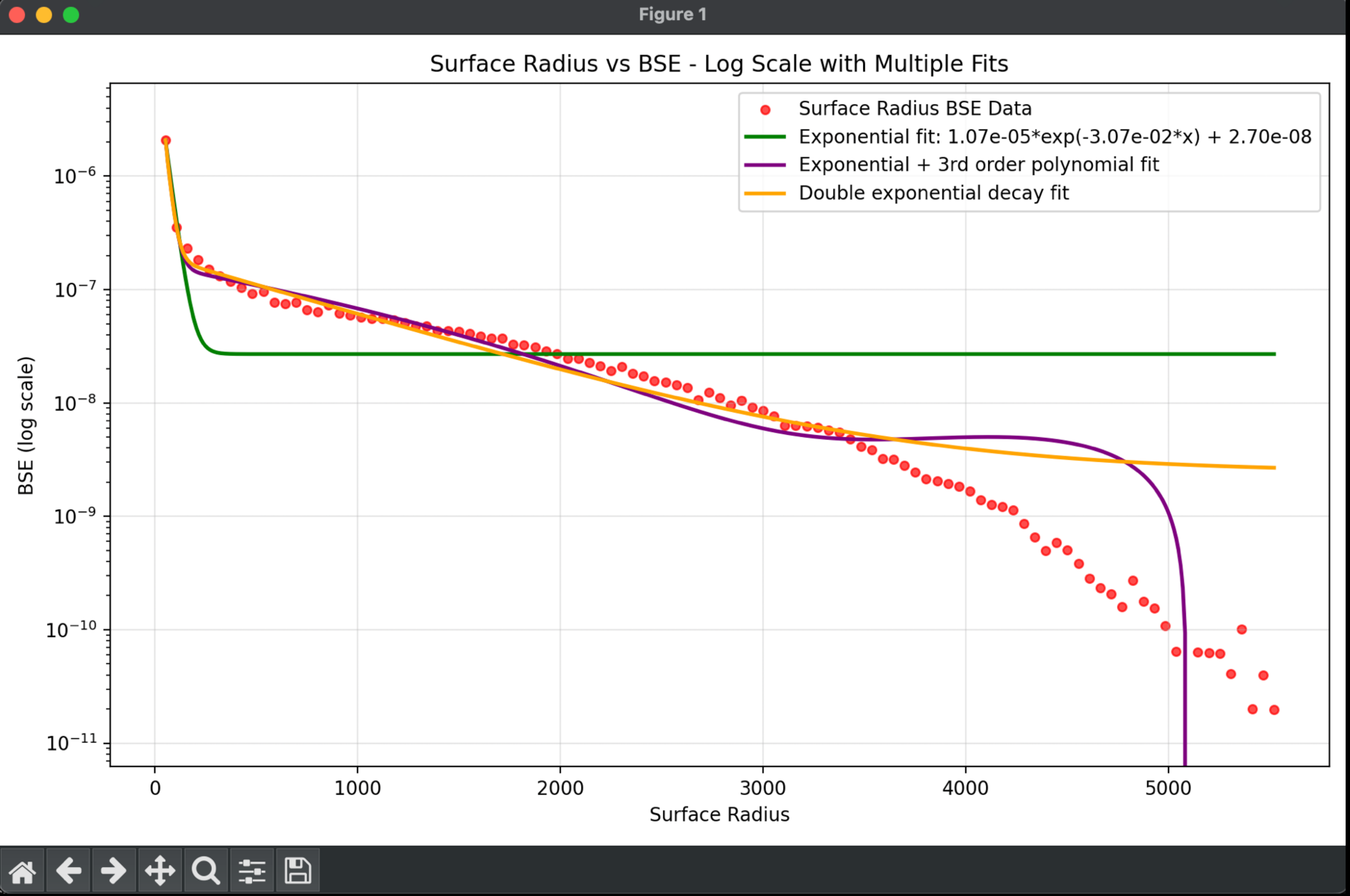
Ln 6, Col 26 Spaces: 4 UTF-8 Python 3.13.5 (base)



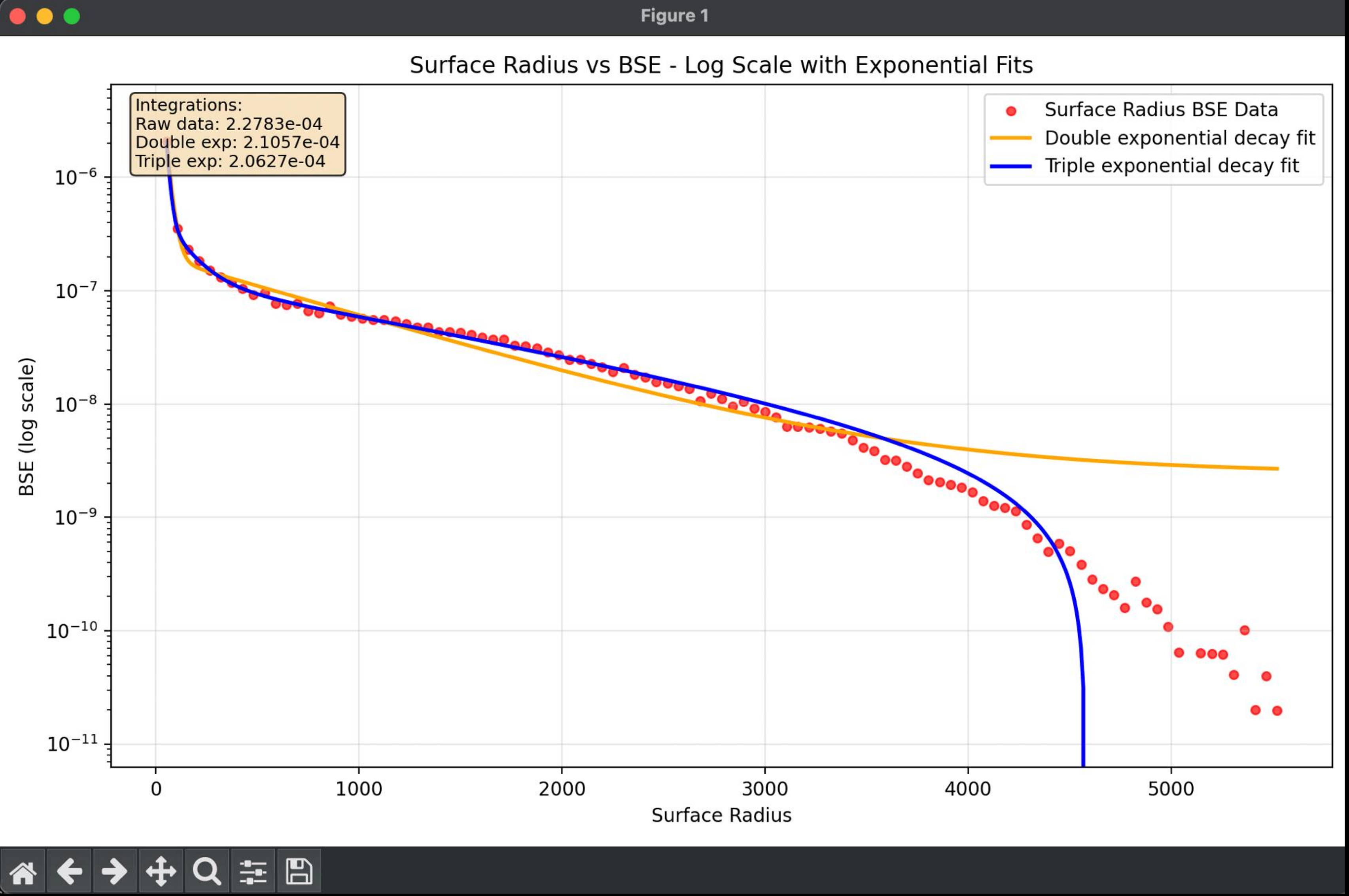
here I tried a simple exponential function fit to the data



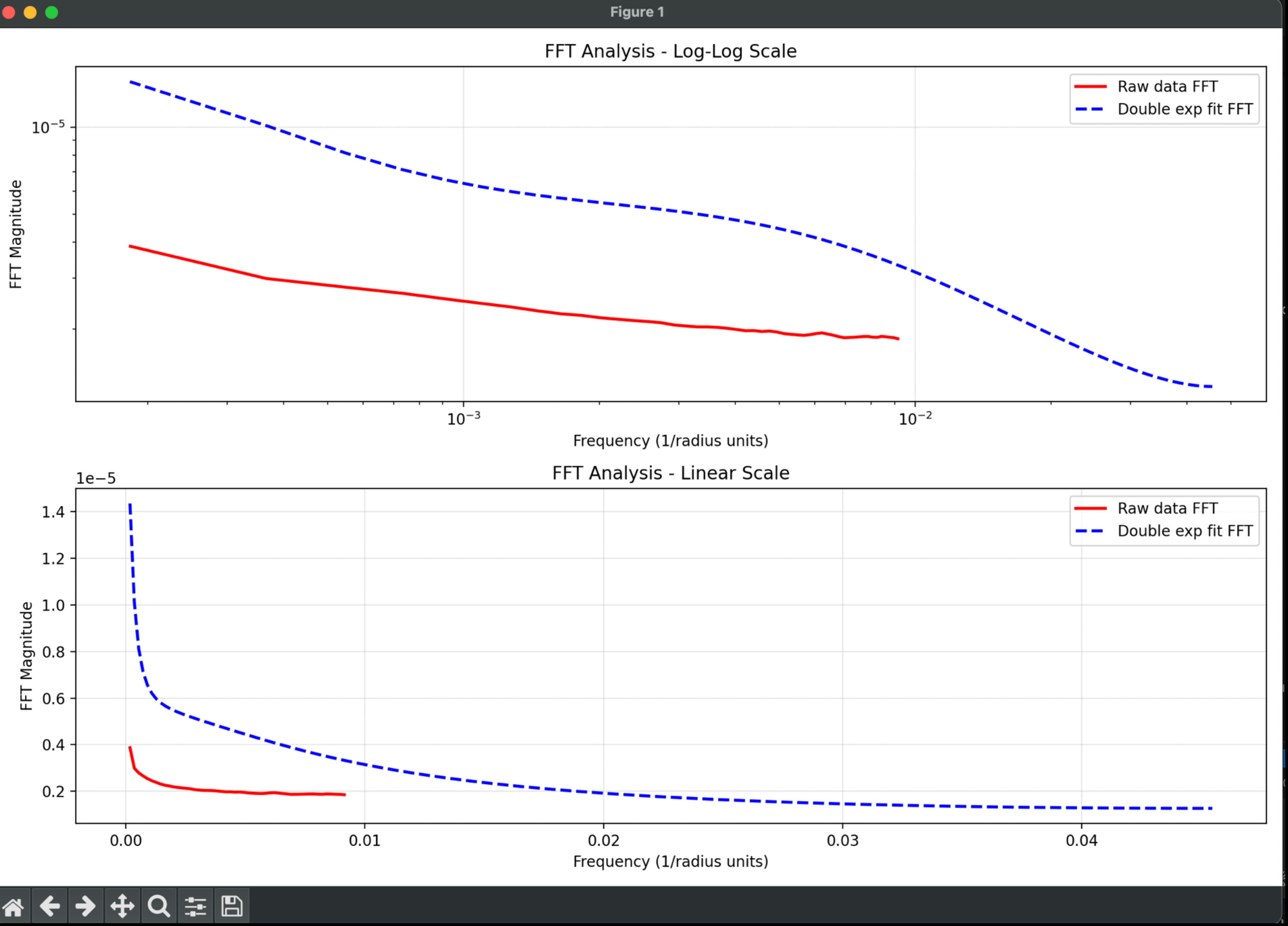
here I tried the sum of an exponential and a polynomial



a few more fits



a couple of fits and some integration results



Fourier Transforms